

SWAPNIL JOSHI

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SUMMARY

Mechanical Engineer with experience in product design, structural analysis, and prototype-to-production transition across research and manufacturing environments. Skilled in CAD modeling, GD&T, tolerance stack-up analysis, FEA validation, and DFM/DFA implementation. Demonstrated ability to reduce rework, improve assembly efficiency, and optimize structural performance through data-driven testing and cross-functional collaboration. Interested in advanced manufacturing and high-performance mechanical systems.

EDUCATION

Clemson University, Clemson, SC

Master of Science in Mechanical Engineering

August 2023 - May 2025

GPA - 3.2/4.0

Savitribai Phule Pune University, Pune, Maharashtra

Bachelor of Engineering in Mechanical Engineering

July 2018 - June 2022

GPA - 3.5/4.0

EXPERIENCE

Research Assistant - Clemson University — Clemson, SC, USA

October 2025 - Present

- Designed and fabricated 15+ auxetic composite specimens via SLA additive manufacturing for stress-sensing applications.
- Conducted static and high-cycle fatigue testing (100,000+ cycles) to validate structural durability.
- Performed FEA in ANSYS to optimize stress distribution, reducing prototype iterations by 30%.
- Developed MATLAB workflows to correlate strain with magnetic sensor output, improving data accuracy by 20%.
- Trained 10+ students in SolidWorks modeling, ANSYS simulation, and additive manufacturing processes.

Product Design and Development Engineer Intern - Manufast.in — Pune, India

July 2022 - June 2023

- Designed 20+ production-ready fixtures and mechanical assemblies using SolidWorks from concept to release.
- Applied GD&T and tolerance stack-up analysis, reducing machining rework by 18%.
- Implemented DFM/DFA strategies, cutting assembly time by 22% and improving build efficiency.
- Resolved prototype fitment issues with CNC and fabrication teams; maintained BOMs and revision control.
- Supported Lean-based process improvements to enhance throughput and reduce waste.

R&D And Digging System Head - Team MechAgrinics — Pune, India

June 2019 - June 2022

- Led development of a hydraulically actuated digging mechanism from concept to field validation.
- Designed structural and linkage assemblies considering load paths, stiffness, and manufacturability.
- Reduced structural weight by 12% through iterative design refinement while maintaining safety factors greater than 2.0.
- Coordinated cross-functional integration with electrical and fabrication teams to deliver production-capable hardware within project deadlines.
- Performed hands-on machining, welding, and system-level debugging during rapid prototyping cycles.

PROJECTS

Suspension and Steering Subsystem Simulation - Clemson University

January 2025 - May 2025

- Developed a front suspension and steering model in Solidworks; verified through kinematic and load transfer analysis.
- Performed DOE-based sensitivity analysis on suspension geometry, improving cornering stability and ride comfort

Lightweight Chassis Design and Optimization Study - Clemson University

August 2024 - December 2024

- Designed a full vehicle frame and optimized structural members for stiffness-to-weight ratio using topology optimization.
- Conducted FEA and modal analysis to validate safety factors and minimize resonance issues.

Autonomous 360-degree Fire Extinguisher Robot - Savitribai Phule Pune University

August 2021 - June 2022

- Built an autonomous fire extinguisher robot that maneuvers in various environments to extinguish fires.
- Created a 360-degree fire suppression system within the robot, enabling it to tackle fires from any angle efficiently and reducing emergency response time by 25%.

Hydraulic Lift and Drop Arm - Savitribai Phule Pune University

August 2018 - June 2019

- Developed a hydraulic lift and drop arm solution to facilitate smoother operations and move heavy loads, reducing manual handling time by 30% and decreasing injury reports among staff.

SKILLS

Software Packages: SolidWorks, Siemens NX, AutoCAD, CATIA, Ansys, Matlab

Manufacturing: CNC Machining, 3D Printing, GD&T, DFM/DFA, FMEA

Programming: C, C++, Python

Optimization and Quality Control: Lean Manufacturing, Six Sigma, Process Optimization, Quality Control Analysis, Efficiency Improvement Techniques, PLM systems, SAP, Minitab

PUBLICATIONS

Panat, Shrirang Sandip, Parth Patil, Swapnil Joshi, and David Gaikwad. "Design and Development of a 360-degree Fire Extinguisher Robot using Microcontroller", 9(5), 547-551, May 2022.